SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – INDIVIDUAL TALK

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 3 – INDIVIDUAL TALK
Weightage:	30 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
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Section / Batch:	2024	**Date:**	27.10.26

Code of Conduct

I _____T BhanuTeja/192472259____ *(Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ____BhanuTeja_____

Instructions

- Topic given before the session and students speak on stage.
- Duration: 10 minutes (In class)

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Individual Task: Location Management Techniques in Mobile Computing

1. Introduction

Location Management is an important process in mobile computing that helps cellular networks identify the approximate location of mobile users. It enables the network to deliver incoming calls, messages, and data services even when users move from one location to another.

In simple words, location management helps the network know which area a mobile phone is currently in so that it can locate the device when communication is required.

2. Basic Concept

Location management can be represented as:

Mobile Phone → Base Station → Location Area → Location Database → Paging → Mobile User

Location management mainly consists of two important processes:

- **Location Registration:** The mobile device updates the network when it enters a new location area or when a location update is required.
- **Location Paging:** The network searches for the mobile device within its registered location area when an incoming call or communication request arrives.

These processes help maintain connectivity while reducing unnecessary network operations.

3. Main Components

1. Mobile Station (MS)

The Mobile Station refers to the mobile device used by the subscriber. It communicates with nearby base stations and provides information needed for network registration and location updates.

2. Base Transceiver Station (BTS)

The BTS provides wireless communication between the mobile phone and the cellular network.

It helps the network identify the cell in which the mobile device is communicating.

3. Base Station Controller (BSC)

The BSC controls multiple base stations in traditional GSM networks. It manages radio resources and supports communication and mobility-related procedures between cells.

4. Home Location Register (HLR)

The HLR is a central database in traditional GSM networks that stores subscriber information, service details, and information about the network currently serving the subscriber.

5. Visitor Location Register (VLR)

The VLR temporarily stores information about subscribers currently present in its service area. It helps the network manage location information and provide services to visiting subscribers.

6. Mobile Switching Center (MSC)

The MSC manages voice calls and coordinates mobility-related procedures in traditional cellular networks. It helps route incoming calls toward the network area where the mobile subscriber is registered.

4. Location Management Techniques

Location management uses the following techniques:

- **Location Registration:** Updates the network when a mobile device enters a new location area or meets specific update conditions.
- **Location Update:** Maintains current location information in the network's databases.
- **Paging:** Searches for a mobile device in a location area when an incoming call or service request arrives.
- **Handoff Management:** Helps maintain an ongoing call or data session when a user moves between cells, subject to network conditions.

These techniques work together to support mobile connectivity and efficient network operation.

5. Location Management in 4G and 5G Networks

4G LTE

In 4G LTE, the network uses Tracking Areas to manage the locations of mobile devices. The Mobility Management Entity (MME) handles tracking-area updates and paging-related control procedures.

5G Networks

In 5G, the Access and Mobility Management Function (AMF) manages registration, mobility, and reachability procedures. The network uses registration areas and paging to locate devices when required.

Modern networks improve mobility management to support large numbers of connected devices and efficient communication.

6. Advantages of Location Management

- Enables incoming calls and messages to reach mobile users.
- Supports mobility when users travel between different network areas.
- Reduces unnecessary searching across the entire network.
- Improves network resource utilization.
- Helps maintain connectivity during movement between cells.

7. Conclusion

To conclude, Location Management Techniques are essential for efficient mobile communication. Location registration and updating help the network maintain information about mobile users, while paging helps locate devices when communication is required. Components such as BTS, MSC, HLR, and VLR support these functions in traditional cellular networks, while 4G and 5G use more advanced mobility management mechanisms. Effective location management improves network efficiency, supports user mobility, and enables reliable delivery of mobile services.

Geotag Photo:

